

SCIENTIFIC ANALYSIS GROUP

Defense Research & Development Organisation

QSAFire: Design and Performance Evaluation of Satellite-Assisted SNS-TF-QKD over the Delhi-Mumbai Quantum Communication Link

Internship Report

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Submitted by

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Certificate

This is to certify that the internship report titled **"QSAFire: Design and Performance Evaluation of Satellite-Assisted SNS-TF-QKD over the Delhi-Mumbai Quantum Communication Link"** is a bona fide record of the work carried out by **Anand Kumar**, a student of Bachelor of Technology (Department of Electronics and Communication Engineering), Bharati Vidyapeeths' College of Engineering, during an internship of duration June- August 2026 at the Scientific Analysis Group, Defence Research and Development Organisation (DRDO), Metcalfe House, Delhi.

The work reported here was carried out under my supervision and, to the best of my knowledge, has not been submitted elsewhere for the award of any other degree or diploma. The report is submitted for evaluation as part of the internship requirements of the student's academic programme.

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I would like to thank Mr. Rajesh Kumar, Scientist ‘E’, Scientific Analysis Group, DRDO, Metcalfe House, for agreeing to supervise this internship and for the direction he gave at points where the work could easily have gone off track. The scope of the three tasks reported here, an OptiSystem prototype, a CSV preprocessing pipeline, and an independent satellite-link simulation study, was shaped substantially by conversations with him, and several of the modelling choices described in later chapters (in particular the decision to keep the physical-layer OptiSystem model and the protocol-layer Python analysis strictly separate) came directly out of that guidance.

I am also grateful to the Scientific Analysis Group at DRDO for the working environment and access to the software needed to carry out this work, and to the scientists and staff at Metcalfe House who answered questions along the way, even when those questions were basic ones.

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Anand Kumar
November 13, 2026

Abstract

This report documents three technical tasks carried out during an internship at the Scientific Analysis Group, DRDO, all concerned with Sending-or-Not-Sending Twin-Field Quantum Key Distribution (SNS-TF-QKD), but executed as separate, largely independent pieces of work rather than stages of one pipeline, and presented here as such.

The first task is an OptiSystem implementation of an SNS-TF-QKD optical link across three stations, Alice, Bob, and an untrusted measurement station Charlie, driven at the physical layer by CW lasers, Mach-Zehnder and intensity modulators, and single-photon detectors, with the protocol-level decision logic (window selection, send/not-send choice, decoy-intensity selection, phase randomisation) implemented as MATLAB co-simulation blocks inside the OptiSystem schematic. It is a laboratory-scale proof of concept (40 km fibre spans, a 10 Gbit/s symbol clock, a 1024-bit sequence per run) with no satellite or orbital modelling. The second task is a Python pipeline that ingests the CSV logs this OptiSystem model exports, aligns them by bit index, cleans formatting inconsistencies, and computes the sifting, gain, and error-rate quantities needed before a secure-key-rate estimate; it operates only on OptiSystem-generated data and is unconnected to the satellite framework below.

The third task, separate from the first two, is a Python satellite-link performance evaluation framework built from first principles: orbital geometry (500 km sun-synchronous, inclination solved from J_2 perturbation theory), elevation-dependent slant range, atmospheric and Rytov-variance turbulence loss, pointing loss, and a fully asymmetric SNS-TF-QKD key-rate calculation, applied to six Indian ground sites and four candidate links. The primary link, Delhi-Mumbai, returns a simulated secure key rate of 0.0947 kbit/s over 1149.4 km, roughly two orders of magnitude below Micius's flagship figures but consistent with real, non-ideal portable ground-station demonstrations. The framework's orbital and loss-budget formulas are cross-checked against Micius's published in-orbit performance (Liao et al., 2017) to within 1 to 3 percent, and the analysis identifies night-sky background light, rather than turbulence, as the dominant constraint on the Delhi-Mumbai link.

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List of Abbreviations

Abbreviation	Expansion
AOD	Aerosol Optical Depth
BB84	Bennett-Brassard 1984 (quantum key distribution protocol)
CSV	Comma-Separated Values
CW	Continuous Wave
DRDO	Defence Research and Development Organisation
FSO	Free-Space Optical
IM	Intensity Modulator
ITU-T	International Telecommunication Union, Telecommunication Standardization Sector
LOS	Line of Sight
MDI-QKD	Measurement-Device-Independent Quantum Key Distribution
MZM	Mach-Zehnder Modulator
NRZ	Non-Return-to-Zero
PM	Phase Modulator
PNS	Photon-Number-Splitting (attack)
PRBS	Pseudo-Random Bit Sequence
QBER	Quantum Bit Error Rate
QKD	Quantum Key Distribution
QSAFire	Quantum Satellite Analysis Framework for Integrated Research and Evaluation
SAG	Scientific Analysis Group
SKC	Secret-Key Capacity
SKR	Secure Key Rate
SMF	Single-Mode Fibre
SNS	Sending-or-Not-Sending
SNS-TF-QKD	Sending-or-Not-Sending Twin-Field Quantum Key Distribution
SPD	Single-Photon Detector
TF-QKD	Twin-Field Quantum Key Distribution
WCP	Weak Coherent Pulse

List of Symbols

Symbol	Meaning
η	Channel transmittance
α	Fibre attenuation coefficient (dB/km)
L	Link distance (km)
ε	Sending probability in the SNS Z-basis
μ_1, μ_2	Signal and decoy mean photon numbers
k	Optical wavenumber, $k = 2\pi/\lambda$
C_n^2	Refractive-index structure parameter
σ_R^2	Rytov variance (scintillation index)
r_0	Fried parameter (atmospheric coherence length)
τ	Atmospheric optical depth
ζ	Zenith angle
e_a	Misalignment error parameter
QBER	Quantum bit error rate (fraction)
R	Secure key rate (bit/s or kbit/s)

Chapter 1

Introduction

1.1 Background and Motivation

Quantum key distribution (QKD) lets two parties agree on a shared secret key whose security rests on the physics of quantum measurement rather than on a computational hardness assumption [1]. This property is attractive for defence and government communication links, but standard point-to-point QKD is bound by a repeaterless rate-distance limit: the achievable key rate falls off exponentially with channel loss, which for fibre means an exponential fall-off with distance. Twin-Field QKD (TF-QKD) breaks this limit without a quantum repeater by having both parties send phase-coherent pulses to a central, untrusted measurement station and using the interference outcome, rather than a direct transmission, to generate correlated bits; this changes the key-rate scaling from linear to square-root in the overall channel transmittance [2]. The Sending-or-Not-Sending (SNS) variant of TF-QKD, used throughout this internship, removes a specific security gap in the original TF-QKD proposal by having each party make a private send/not-send decision for every key-generating time window instead of publicly disclosing phase information for those windows [3]. SNS-TF-QKD also tolerates a much larger single-photon misalignment error than the original scheme, which matters directly for any free-space or satellite link, where interference visibility is harder to maintain than in fibre.

This background is stated briefly here because it recurs, in different forms, across all three tasks reported in this document: Task 1 builds a physical-layer SNS-TF-QKD link in OptiSystem, Task 2 processes the CSV output of that link, and Task 3, the QSAFire framework, evaluates whether an SNS-TF-QKD link of this kind is plausible over a satellite downlink to Indian ground stations. None of the three tasks requires the reader to be re-taught QKD theory from first principles, and this report does not attempt to; the relevant protocol detail is introduced again, briefly, at the start of each task-specific chapter, only where it affects an implementation choice.

1.2 Scope of the Internship

The internship consisted of three technical tasks. They were carried out as separate pieces of work, using separate tools, and they are reported separately in this document. No

output of Task 1 or Task 2 feeds into Task 3, and no attempt has been made here to present them as stages of one pipeline, because they are not.

Task 1: OptiSystem SNS-TF-QKD Prototype

A physical-layer proof-of-concept implementation of an SNS-TF-QKD link in OptiSystem, with the protocol-level decision logic (window selection, send/not-send choice, decoy-intensity selection, phase randomisation) implemented through embedded MATLAB co-simulation blocks. The model uses fibre spans (SMF-28) between three stations, Alice, Bob and Charlie, at a laboratory-representative bit rate, and does not model a satellite channel or orbital geometry of any kind. Chapter 2 covers this task.

Task 2: Python Preprocessing Pipeline

A pipeline that reads the CSV files exported by the Task 1 OptiSystem model, aligns them by bit index, cleans up export artefacts, and computes the sifting, gain and error-rate quantities needed before a secure-key-rate estimate can be attempted. This pipeline consumes only OptiSystem-generated data. Chapter 3 covers this task.

Task 3: QSAFire

An independent Python simulation framework, QSAFire (Quantum Satellite Analysis Framework for Integrated Research and Evaluation), built to evaluate SNS-TF-QKD performance over a satellite-to-ground downlink. QSAFire models orbital geometry, ground-station visibility, atmospheric and turbulence loss, and a fully asymmetric SNS-TF-QKD secure-key-rate calculation, applied primarily to the Delhi-Mumbai ground-station pair and, secondarily, to three other Indian city pairs. QSAFire shares no code and no data path with Task 1 or Task 2. Chapter 4 covers this task in depth, and Chapter 5 carries its results forward into a comparison against the real Micius satellite-QKD mission.

Figure 1.1 summarises this separation. The three boxes are drawn with no connecting arrows between them for a reason: there is no data flow between them, and this report is organised to keep that boundary visible rather than to paper over it.

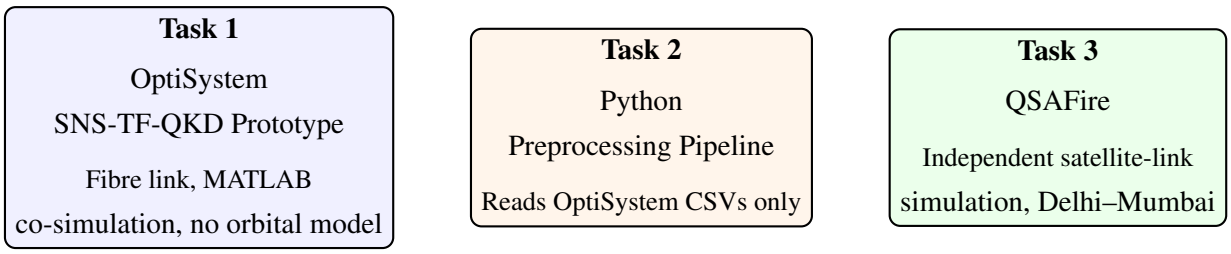


Figure 1.1: The three internship tasks and their independence. Task 1 and Task 2 form one path (OptiSystem output consumed by the Python preprocessing pipeline); Task 3 (QSAFire) is a separate, self-contained simulation study.

1.3 Objectives

The objectives set for the internship, task by task, were:

- Build and simulate a working SNS-TF-QKD link in OptiSystem at the physical layer, including MATLAB-driven protocol logic, and produce synchronised CSV logs from all three stations (Task 1).
- Write a Python pipeline capable of ingesting those CSV logs, correcting for common export issues, and computing the classical pre-processing quantities (sifted key length, gain, raw QBER) needed by a downstream key-rate calculation (Task 2).
- Independently evaluate, in software, whether an SNS-TF-QKD downlink between a low-Earth-orbit satellite and pairs of Indian ground stations is plausible under realistic atmospheric and turbulence conditions, and benchmark the resulting model against the published performance of the Micius satellite, the only satellite-QKD mission with public, peer-reviewed in-orbit data (Task 3, QSAFire).

1.4 Organisation of the Report

Chapter 2 documents the OptiSystem prototype: station-level architecture, MATLAB component roles, and simulation parameters. Chapter 3 documents the Python preprocessing pipeline built around that prototype's CSV output. Chapter 4 documents QSAFire: its orbital, atmospheric, and turbulence models, the ground sites and links it was applied to, and a specific correction made to its turbulence model partway through the internship, together with the verified effect of that correction on the final results. Chapter 5 draws together QSAFire's results across all four ground-station links and benchmarks the Delhi-Mumbai link against Micius. Chapter 7 closes the report with

an evidence-based summary of what was completed, its limitations, and a short account of reasonable next steps for each task.

1.5 Chapter Summary

This internship comprised three independent tasks under a shared broad theme, SNS-TF-QKD, but different scopes, tools, and levels of physical versus simulated implementation: an OptiSystem physical-layer prototype (Task 1), a Python CSV preprocessing pipeline for that prototype (Task 2), and QSAFire, an independent satellite-link simulation framework (Task 3). The remainder of this report treats each on its own terms and does not present them as one integrated system.